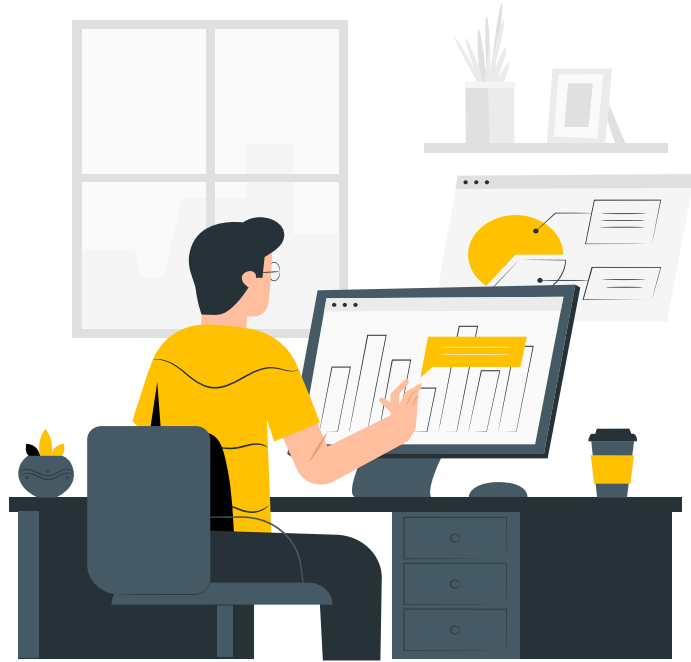


Statistika Sosial

#14-15 Meeting

Korelasi dan Regresi Linear Sederhana

Ferdian Bangkit Wijaya, S.Stat., M.Si
Agung Satrio Wicaksono, S.Mat., M.Si



ANALISIS KORELASI DAN REGRESI LINIER SEDERHANA

PENDAHULUAN

- Kita umumnya ingin mengetahui hubungan antar peubah.
- **Analisis Korelasi** digunakan untuk melihat keeratan hubungan linier antar dua peubah.
- **Analisis Regresi** digunakan untuk melihat hubungan sebab-akibat antar peubah.



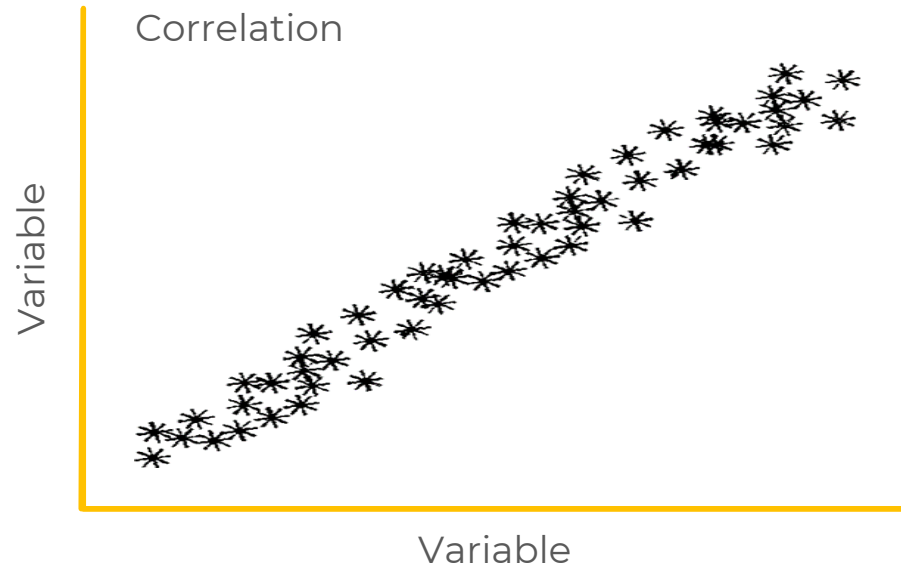
ASSOCIATION



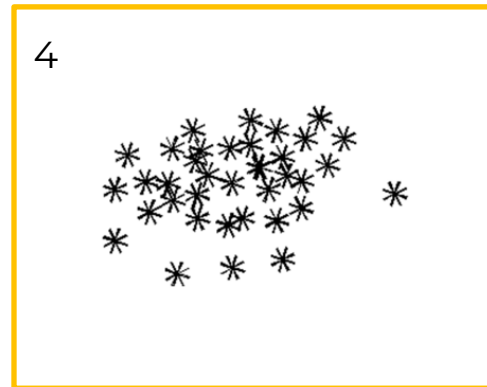
CORRELATION

summarizes the strength and direction of the linear association between two quantitative variables.

OVERVIEW



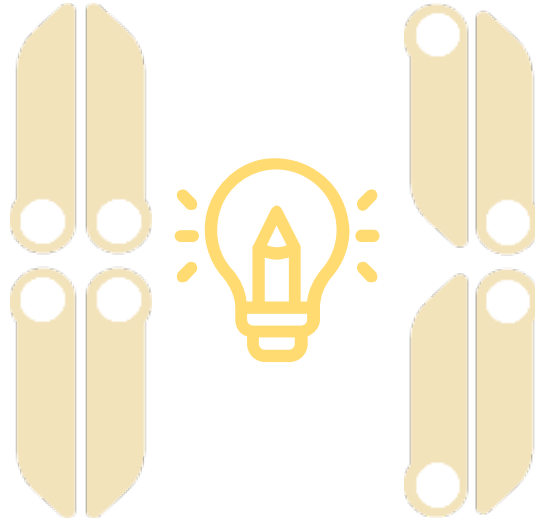
ASSOCIATION BETWEEN TWO VARIABLES



POSITIVE AND NEGATIVE ASSOCIATION

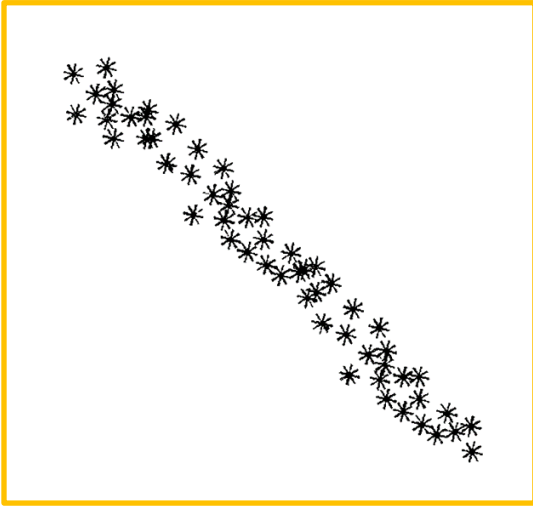
Two quantitative variables x and y are said to:

have a **positive association**
when x goes up
and y tends to go
up; or when x
goes down and y
tends to go
down.



have a **negative association**
when x goes up,
and y tends to go
down, or vice
versa.

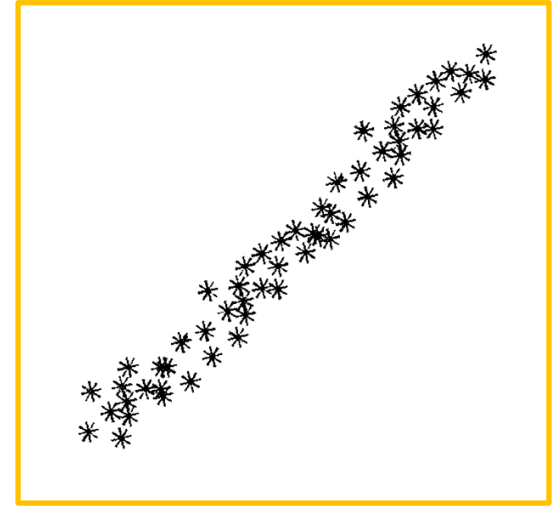
CORRELATION



Negative



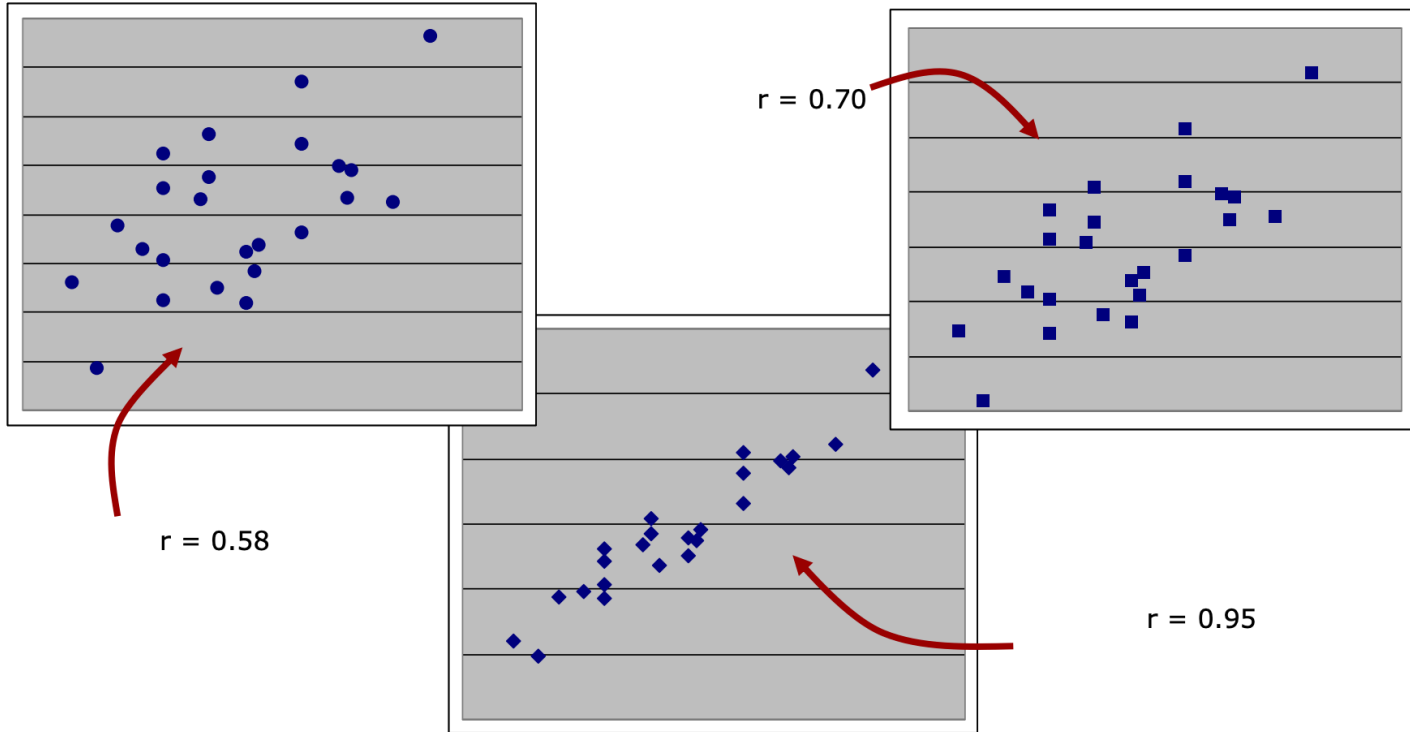
Zero



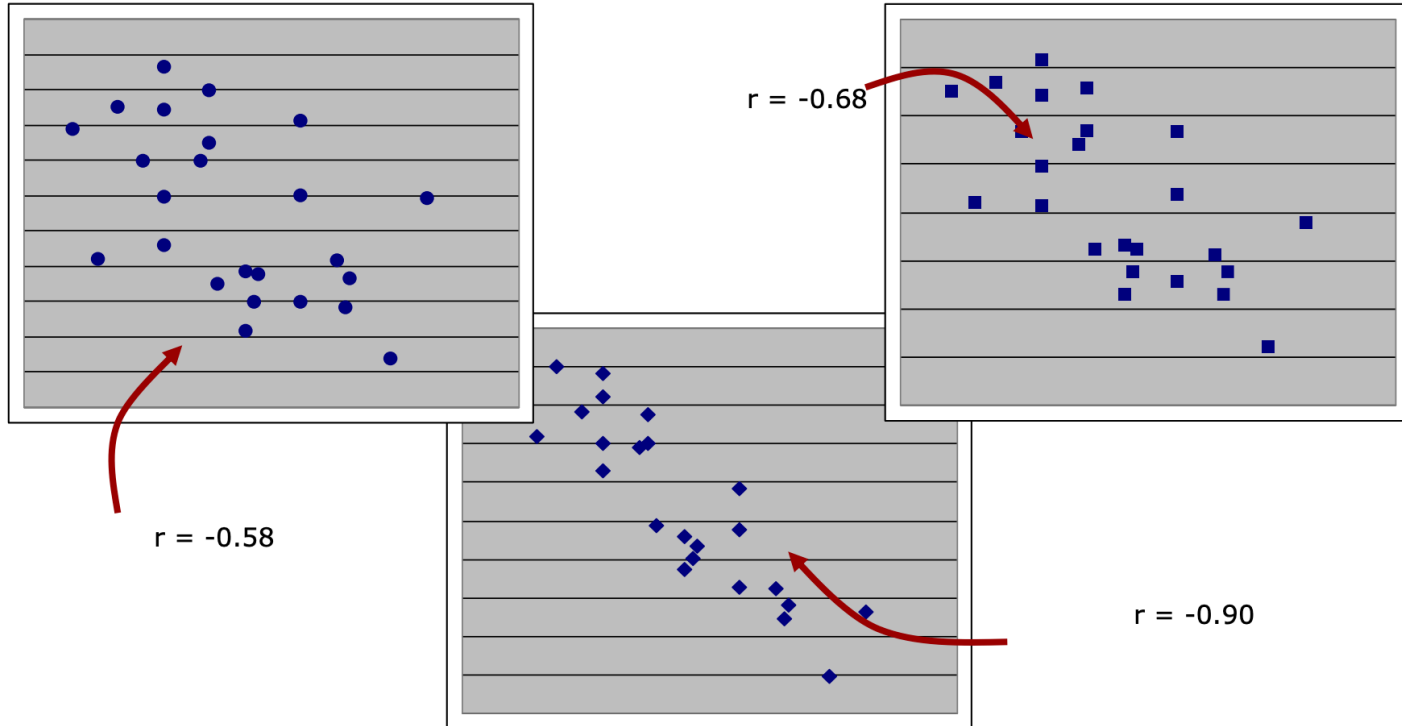
Positive



POSITIVE CORRELATION



NEGATIVE CORRELATION



CALCULATING THE CORRELATION

$$r_{xy} = \frac{s_{xy}}{\sqrt{s_{xx}}\sqrt{s_{yy}}}$$

Where:

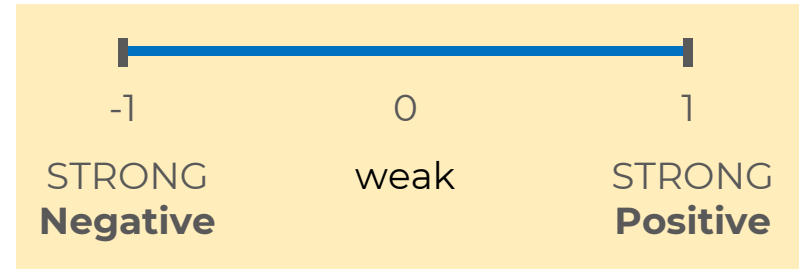
- $s_{xy} = n \sum xy - (\sum x)(\sum y)$
- $s_{xx} = n \sum x^2 - (\sum x)^2$
- $s_{yy} = n \sum y^2 - (\sum y)^2$

r = correlation

s = standard deviation

n = number of observation

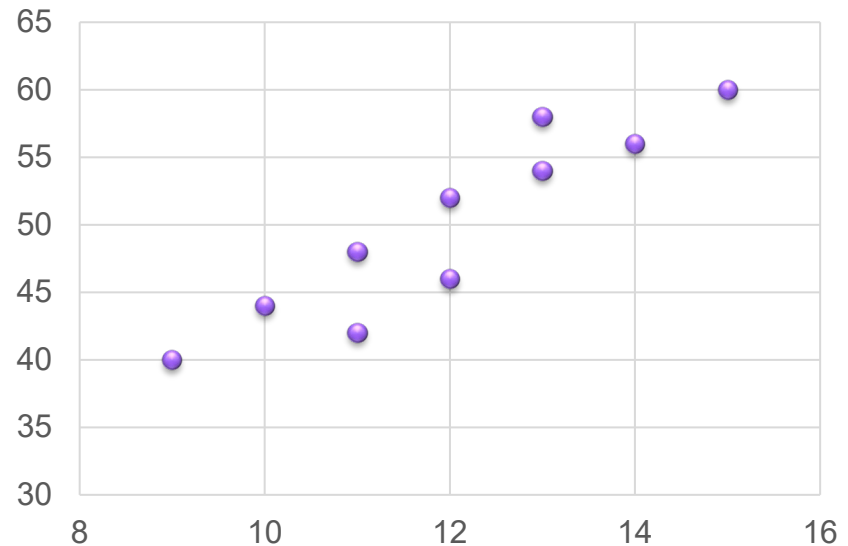
Correlation Coefficient:



ILUSTRASI

- Ingin dilihat hubungan antara pengeluaran iklan dan banyaknya penjualan

Cabang	Iklan (juta rupiah)	Banyaknya penjualan (unit)
1	10	44
2	9	40
3	11	42
4	12	46
5	11	48
6	12	52
7	13	54
8	13	58
9	14	56
10	15	60



ILUSTRASI

X	Y	XY	X ²	Y ²
10	44	440	100	1936
9	40	360	81	1600
11	42	462	121	1764
12	46	552	144	2116
11	48	528	121	2304
12	52	624	144	2704
13	54	702	169	2916
13	58	754	169	3364
14	56	784	196	3136
15	60	900	225	3600
120	500	6106	1470	25440

Didapatkan:

- $s_{xy} = n \sum xy - (\sum x)(\sum y) = 1060$
- $s_{xx} = n \sum x^2 - (\sum x)^2 = 300$
- $s_{yy} = n \sum y^2 - (\sum y)^2 = 4400$

Maka:

$$r_{xy} = \frac{s_{xy}}{\sqrt{s_{xx}}\sqrt{s_{yy}}} = \frac{1060}{\sqrt{300}\sqrt{4400}} = 0.9226$$

Nilai tersebut menunjukkan adanya hubungan positif yang kuat antara pengeluaran iklan dengan banyaknya penjualan.



PENYALAHGUNAAN KOEFISIEN KORELASI

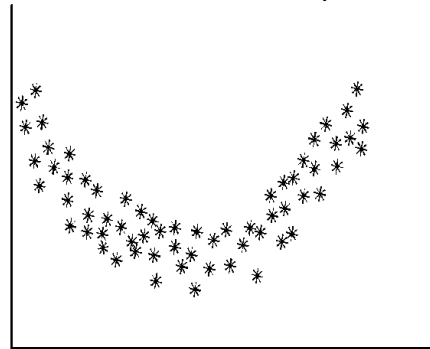
A strong correlation **does not mean** a cause-and-effect relationship. Causation is **only one explanation** of an observed association.

Strong correlation does not mean



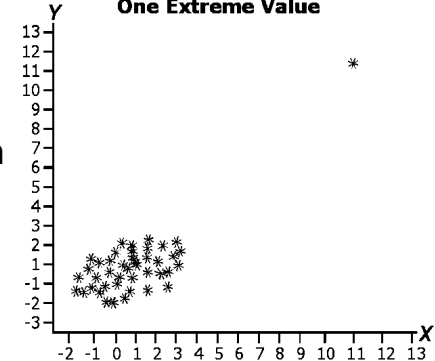
Missing another type of relationship

Curvilinear Relationship



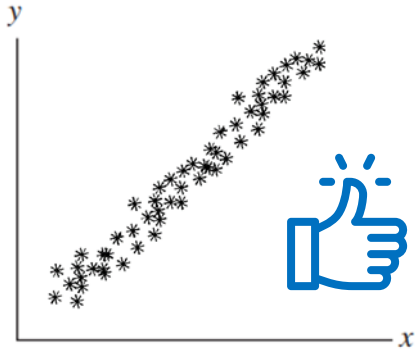
Extreme data values

Correlation with One Extreme Value

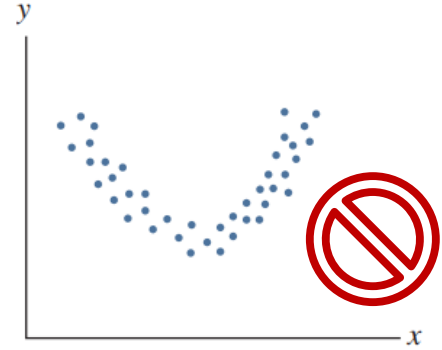


IN PRACTICE: ALWAYS CONSTRUCT A SCATTERPLOT

Always construct a scatterplot to display a relationship between two quantitative variables



The correlation is only meaningful for describing **the direction** and **strength** of an approximate **straight-line association**.



REGRESSION

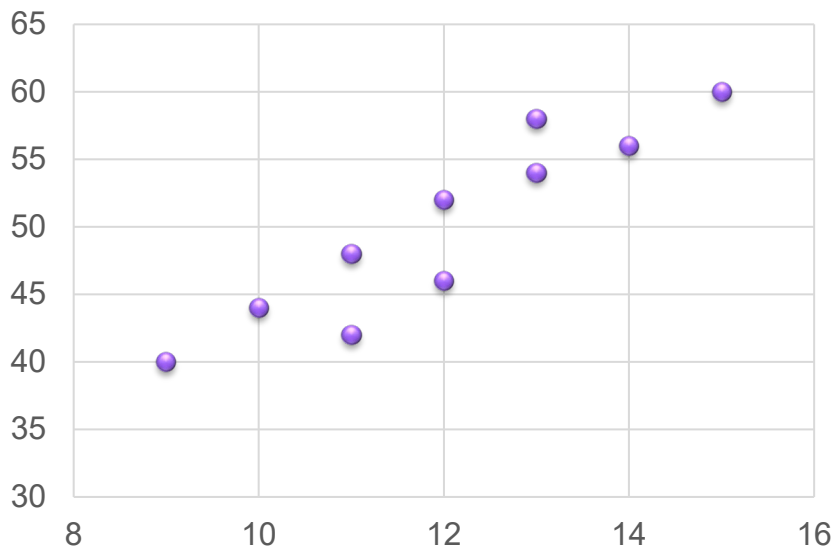
concerns the **study of relationships** between quantitative variables with the object of identifying, estimating, and validating the relationship.

The estimated relationship can then be used to predict one variable from the value of the other variable(s).

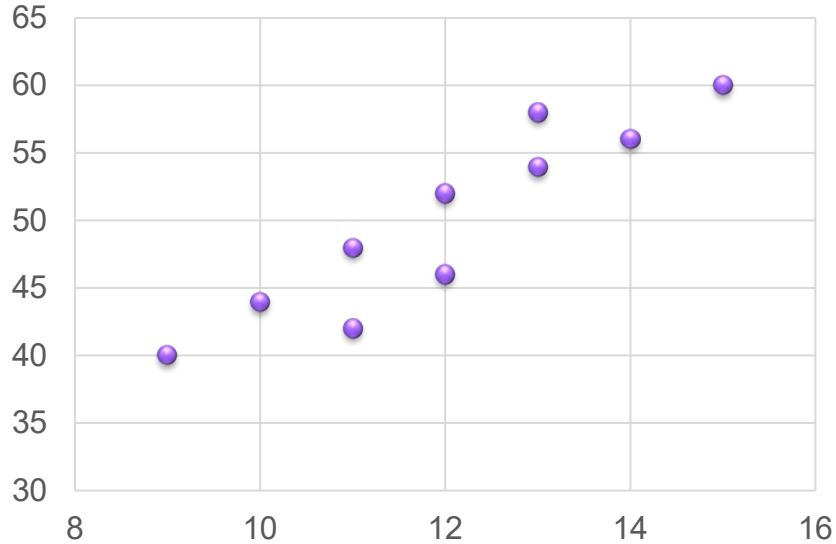
ILUSTRASI

- Seberapa efektif iklan yang dipasang dalam mendorong penjualan?
- Dilihat catatan belanja iklan dan penjualan dari beberapa toko cabang

Cabang	Iklan (juta rupiah)	Banyaknya penjualan (unit)
1	10	44
2	9	40
3	11	42
4	12	46
5	11	48
6	12	52
7	13	54
8	13	58
9	14	56
10	15	60



Banyaknya
penjualan (unit)



Pengeluaran untuk
iklan (juta rupiah)

ILUSTRASI

Regresi Linier

mencari garis lurus

$$\hat{Y} = \alpha + \beta X$$

yang paling menggambarkan pola hubungan antara kedua peubah tsb.

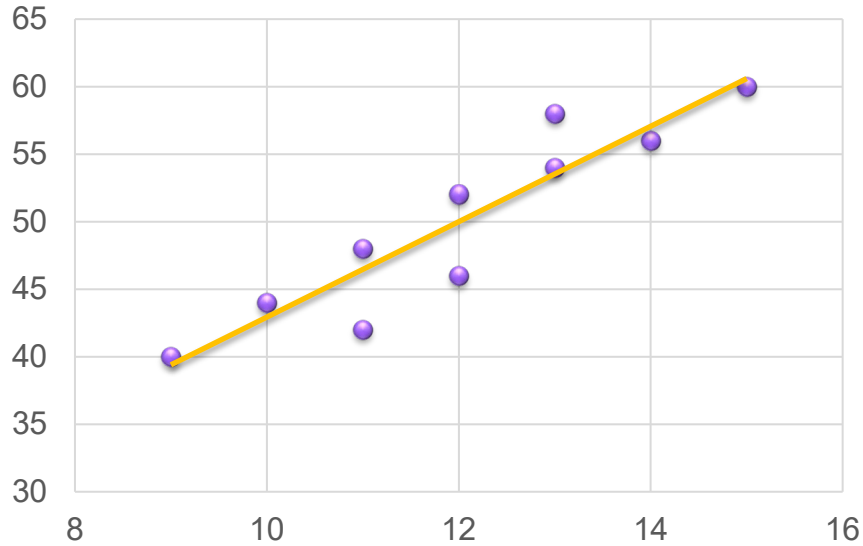
x = peubah bebas / peubah penjelas /
peubah prediktor

y = peubah tak bebas / peubah terikat /
peubah respon



Banyaknya
penjualan (unit)

ILUSTRASI



Pengeluaran untuk
iklan (juta rupiah)

Ingin dibuat model

$$\hat{Y} = \alpha + \beta X$$

Model memuat error, selisih **nilai sebenarnya** dengan **dugaan berdasar model**

$$e = Y - \hat{Y}$$



BAGAIMANA MENDAPATKAN α DAN β ?

- **Metode Kuadrat Terkecil** (*Ordinary Least Square*, OLS) digunakan untuk mencari α dan β sehingga didapat jumlah kuadrat error paling kecil.
- Cari penduga α dan β sehingga:

$$\min \sum_{i=1}^n e_i^2 = \sum_{i=1}^n [Y_i - \alpha - \beta X_i]^2$$

- dengan

$$\beta = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

- dan

$$\alpha = \bar{Y} - \beta \bar{X}$$



ILUSTRASI PERHITUNGAN

X	Y	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
10	44	-2	-6	12	4
9	40	-3	-10	30	9
11	42	-1	-8	8	1
12	46	0	-4	0	0
11	48	-1	-2	2	1
12	52	0	2	0	0
13	54	1	4	4	1
13	58	1	8	8	1
14	56	2	6	12	4
15	60	3	10	30	9
		Jumlah		106	30

Didapatkan:

$$\bar{X} = 12, \quad \bar{Y} = 50$$

$$\sum_{i=1}^{10} (X_i - \bar{X})(Y_i - \bar{Y}) = 106$$

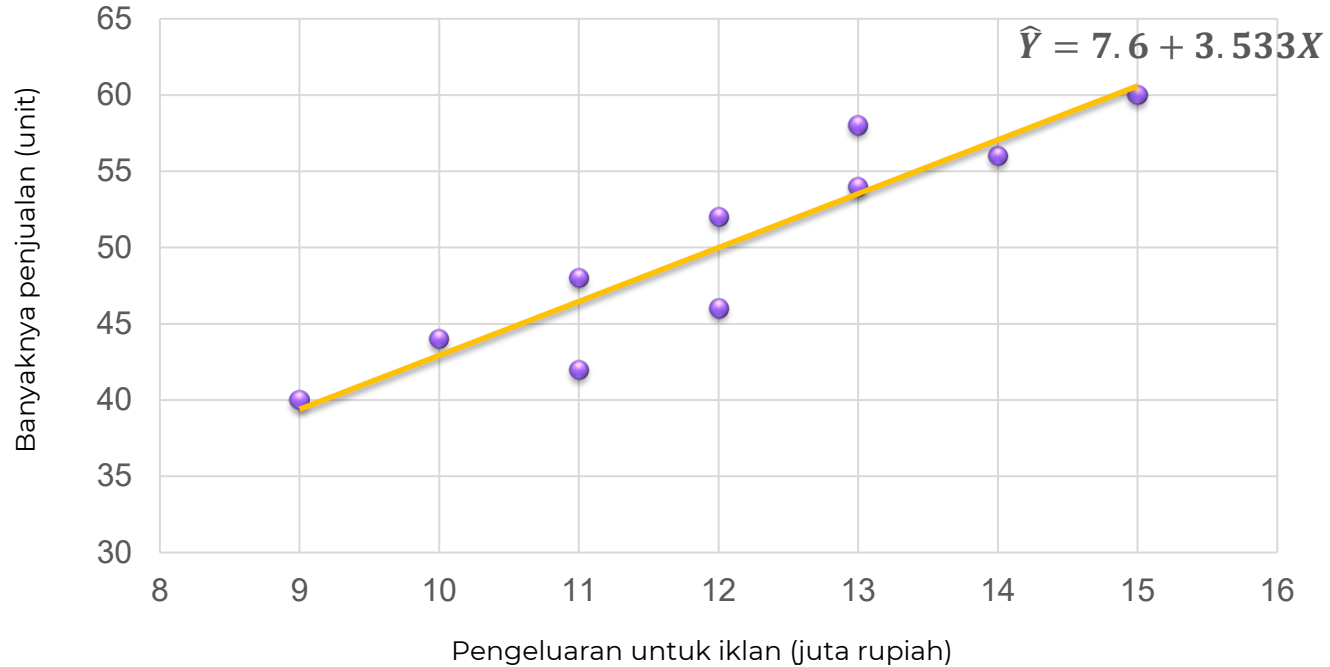
$$\sum_{i=1}^{10} (X_i - \bar{X})^2 = 30$$

$$\beta = \frac{106}{30} = 3.533$$

$$\alpha = 50 - (3.533)12 = 7.6$$



ILUSTRASI



INTERPRETASI PARAMETER REGRESI

Model Regresi:

$$\hat{Y} = \alpha + \beta x$$

di mana:

α : **Intercept** → Besarnya nilai Y pada saat X bernilai 0 (konstan)

β : **Slope** → Besarnya perubahan Y apabila X berubah sebesar satu satuan unit. Tanda koefisien β menunjukkan arah hubungan X dan Y

Pada kasus ilustrasi tsb:

α : **7.6** → Besarnya penjualan jika tidak ada pengeluaran untuk iklan adalah 7.6 unit

β : **3.533** → Jika pengeluaran untuk iklan dinaikkan 1 juta rupiah, maka besarnya penjualan naik sebesar 3.533 unit.



UJI TERHADAP KOEFISIEN β

Perumusan Hipotesis

$H_0: \beta = 0$ (artinya X tidak mempengaruhi Y)

$H_1: \beta \neq 0$ (artinya X mempengaruhi Y)

Statistik Uji

$$t_h = \frac{\beta - 0}{s_b}, \quad \text{dengan } s_b = \sqrt{\frac{\sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{(n-2) \sum_{i=1}^n (X_i - \bar{X})^2}}$$

Kriteria Pengujian

Tolak H_0 jika nilai $t_h > t_{(\frac{\alpha}{2}; db=n-2)}$



UJI TERHADAP KOEFISIEN β

- Nilai $s_b = 0.52$
- Nilai $t_h = 6.765$
- Nilai $t_{(\frac{\alpha}{2}; db=n-2)}$ dengan $\alpha = 0.05, db = 10 - 2 = 8$ yaitu 2.306
- Tolak H_0 jika nilai $t_h > 2.306$

Kesimpulan: Karena $6.765 > 2.306$, maka H_0 ditolak. Artinya terdapat cukup bukti untuk menyatakan bahwa pengeluaran untuk iklan mempengaruhi banyaknya penjualan.



UKURAN KEBAIKAN MODEL

- Menggunakan **Koefisien Determinasi (*R-Square*, R^2)**

$$R^2 = \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

- R-Square (R^2)** merupakan persentase keragaman data yang mampu diterangkan oleh model.
- R^2 bernilai antara 0 sampai dengan 1.
- Semakin tinggi R^2 mengindikasikan semakin baik model.



UKURAN KEBAIKAN MODEL

Pada ilustrasi kasus sebelumnya, didapatkan model:

$$\hat{Y} = 7.6 + 3.533X$$

dengan nilai **R^2 sebesar 0.85.**

Artinya **85% keragaman dari data** mampu diterangkan oleh model, sedangkan sisanya diterangkan oleh hal lainnya yang tidak terdapat dalam model.



MENGGUNAKAN MS. EXCEL

Dengan menggunakan bantuan Microsoft Excel, model regresi dapat diketahui. Langkah-langkahnya adalah:

1. Input data ke Excel
2. Data → Data Analysis → Regression → OK.

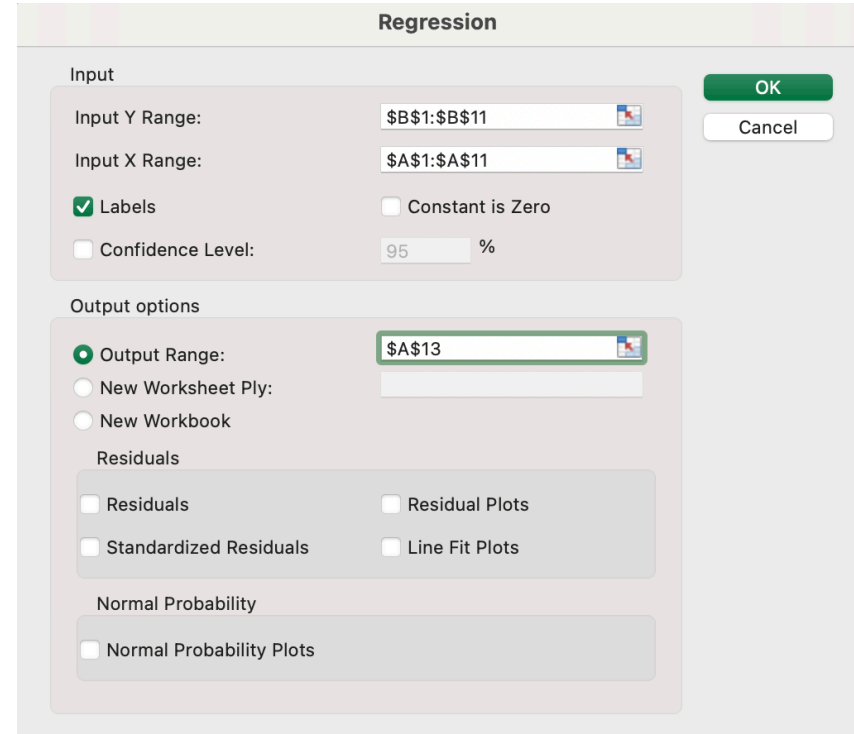
The screenshot displays an Excel spreadsheet with two columns, A and B, containing data for a regression analysis. Column A is labeled 'X' and Column B is labeled 'Y'. The data points are as follows:

X	Y
10	44
9	40
11	42
12	46
11	48
12	52
13	54
13	58
14	56
15	60

Below the spreadsheet, the 'Data Analysis' dialog box is open. The 'Analysis Tools' list includes 'Rank and Percentile', 'Regression', 'Sampling', 't-Test: Paired Two Sample for Means', 't-Test: Two-Sample Assuming Equal Variances', and 't-Test: Two-Sample Assuming Unequal Variances'. The 'Regression' option is selected and highlighted in green. The 'OK' button is visible on the right side of the dialog box.

MENGGUNAKAN MS. EXCEL

3. Isi 'Input Y Range' dengan kolom **peubah respon**, dan 'Input X Range' dengan kolom **peubah bebas** (jika nama peubahnya disertakan dalam input range, jangan lupa centang 'Labels')
4. Pilih 'Output Range' pada cell yang diinginkan. Lalu Klik OK.



The image shows the 'Regression' dialog box in Microsoft Excel. The 'Input' section contains 'Input Y Range' set to '\$B\$1:\$B\$11' and 'Input X Range' set to '\$A\$1:\$A\$11'. The 'Labels' checkbox is checked, and 'Constant is Zero' is unchecked. The 'Confidence Level' is set to 95%. The 'Output options' section has 'Output Range' selected with the value '\$A\$13'. Below this, the 'Residuals' section has 'Residuals', 'Standardized Residuals', 'Residual Plots', and 'Line Fit Plots' all unchecked. The 'Normal Probability' section has 'Normal Probability Plots' unchecked. 'OK' and 'Cancel' buttons are on the right.

Regression	
Input	
Input Y Range:	\$B\$1:\$B\$11
Input X Range:	\$A\$1:\$A\$11
☒ Labels	☐ Constant is Zero
☐ Confidence Level:	95 %
Output options	
☒ Output Range:	\$A\$13
☐ New Worksheet Ply:	
☐ New Workbook	
Residuals	
☐ Residuals	☐ Residual Plots
☐ Standardized Residuals	☐ Line Fit Plots
Normal Probability	
☐ Normal Probability Plots	

MENGGUNAKAN MS. EXCEL

Akan muncul tampilan seperti berikut.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.922611577							
R Square	0.851212121							
Adjusted R Square	0.832613636							
Standard Error	2.860652606							
Observations	10							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	374.5333333	374.533333	45.7678208	0.000142813			
Residual	8	65.46666667	8.18333333					
Total	9	440						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	7.6	6.332324481	1.20019118	0.26439691	-7.002366439	22.2023664	-7.0023664	22.2023664
X	3.533333333	0.522281321	6.76519185	0.00014281	2.328950448	4.73771622	2.32895045	4.73771622

Nilai korelasi → Multiple R (0.9226)

Model regresi → *Coefficients* ($\hat{Y} = 7.6 + 3.533X$)

t hitung → t Stat X (6.765)

Koefisien determinasi → R Square (0.85)



SEE YOU NEXT WEEK !

Ferdian Bangkit Wijaya, S.Stat., M.Si
NIP. 199005202024061001
ferdian.bangkit@untirta.ac.id